Name: _____ School: _____

Name: _____ Team Number: _____ Score: _____/100

2020 UT Austin Science Olympiad Regional Tournament

Water Quality B/C

Directions: You and your partner have 50 minutes to complete this test and test your salinometer at the salinometer station. You may have one sheet of notes and two non-programmable, non-graphing calculators between you and your partner. You must wear eye protection for the salinometer testing, and you may leave the event room to get eye protection if needed as long as the event has not yet started. Please be respectful of your peers at all times. If you have a question, raise your hand at your seat; do not get up.

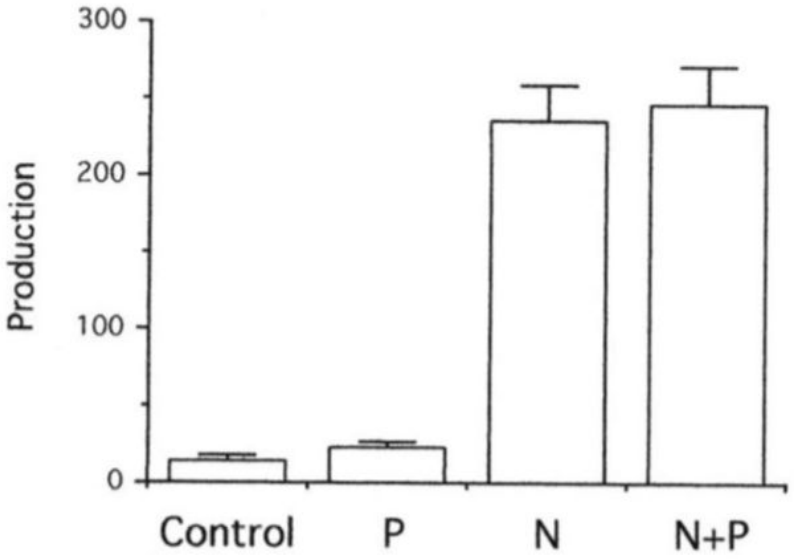
A. Free Response

1. (3) Explain what occurred at Joint Base Water Treatment Facility in Cape Cod, Massachusetts, due to effluent release
2. (1) (T/F) A lentic body of water will have a higher productivity than a lotic body of water?
3. (1) The EPA creates a Contaminant Candidate List every _____ years

4. (8) (**DIVISION C**) The rate constant of the pesticide permethrin in a water column is approximately 0.025 $hour^{-1}$
    a. If the initial concentration of permethrin in a water column is 0.60 µg/mL, what will the concentration be after two days? (3)
    b. Would you expect the rate constant for permethrin to be lower or higher in soil? Why? (3)
    c. Permethrin is not teratogenic but is highly toxic to aquatic organisms. What type of pesticide would the EPA classify permethrin as? (2)

5. In a theoretical experiment, four water samples are taken from a pond and their levels of productivity is measured over one week. In the control condition, no nutrients are added to the samples labeled “N”, only nitrogen is added. In the sample labeled “N+P”, both nitrogen and phosphorus is added. The graph below shows data from such an experiment.
    a. (2) Which nutrient, nitrogen or phosphorus is the limiting nutrient in the pond?

    b. (3) Why is the identity of a limiting nutrient important information for water resource managers?

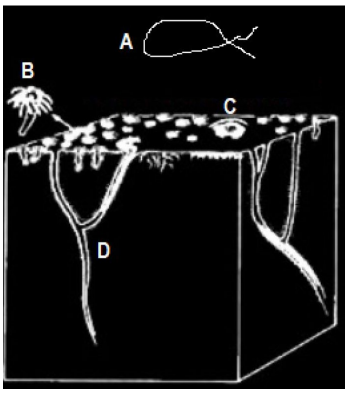


6. (2) Which of the following is/are NOT effective in desalinating water?
    a. Membrane separation
    b. Reverse osmosis
    c. Multi-stage flash distillation
    d. Thermal desorption
    e. Electrodialysis reversal
7. The following questions are related to acid rain.
    a. Why do lakes have a natural buffering capacity? Write the relevant chemical equation (2)
    b. Write the relevant chemical reaction(s) when a strong acid reacts with the buffer (1)
    c. Write the chemical reaction for the production of carbonic acid in water. (1)

8. (1 point each) In order to minimize issues that occur during operations in a potable water treatment facility, a manager has to balance a variety of factors. Match the factor (answer choices) with the problems that arise from them. Answers can be used once, more than once, or not at all.

a. _____ Lowest pH	A. Poor primary classification
b. _____ Rising Sludge	B. Hydraulic overload
c. _____ Highest Energy use	C. Nitrification
d. _____ Lowest DO levels	D. Organic overload
e. _____ Standing water	E. Organic underload
f. _____ Poorest effluent quality	F. Nutrient shortage
g. _____ Plugging	G. Cold Weather

9. (3) When Nitrogen is removed in the process of air stripping, _____ is converted to _____ by raising the pH of the water to _____.

10. 

Observe a section of the abyssal plain. Here you can see a cusk eel (a), an abyssal sponge (c), and a marine isopod (b) and a Zenophophore (d). Label each as demersal, infaunal, mobile epifaunal and fixed epifaunal.

a.

b.

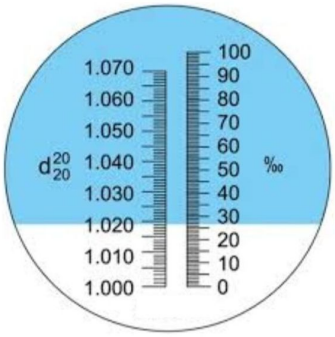
c.

d.

11. (3) Describe the effects of coccolithophorids (a type of phytoplankton) on the global sulfur cycle and the reduction of global warming?

12. (3) Hows does the presence of limestone rock ($CaCO_3$) in and around a body of water affect the water's pH?

13.

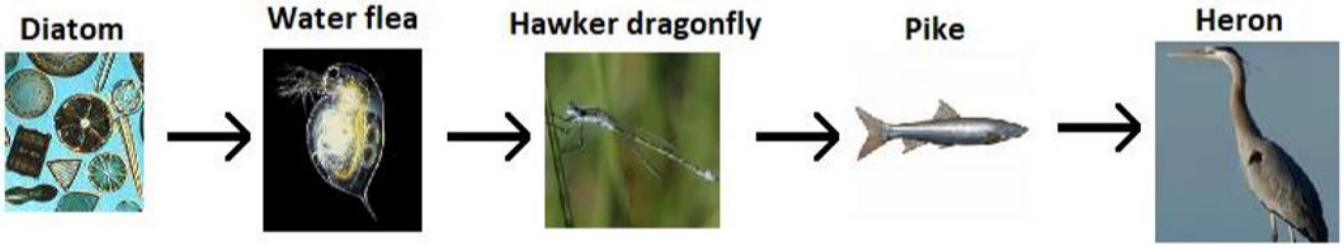


a. (3) What instrument gave this reading?

b. (1) Using the instrument, give the specific density of water?

c. (2) Would this water be classified as Hyperhaline, Matahaline, Ehaline, Polyhaline, Meshaline, or Oligohaline?

14. Consider the following food chain.



a. (2) Based on its position in this chain, a pike is a _____ order consumer

b. (3) A heron has a mass of 3 kilograms. Approximately how much biomass of diatoms is required to sustain one heron?

15. (3) Order the steps for sludge digestion: dewatering, methanogenesis, acetogenesis, acidogenesis, hydrolysis

B. Identification : Refer to the Color Sheet

1. (DIVISION C)

a. Identify the organism (1 point)

b. The organism is in its _____ form. (1 point)

c. (T/F) This species larvae directly released by the female into the water column (2 points)

d. What are the feeding habits of this organism? (2 points)

2.

a. What is the common name of this organism? (1 point)

b. (**DIVISION C**) Briefly describe and identify the two stages of development of this organism. Why? (2 points)

c. (T/F) This organism is an ocean dweller, and only dwells in the shallow part of the ocean (1 point)

d. Describe the defense adaptations of this organism? Is it able to regenerate body parts? Explain briefly. (3 points)

3.

a. Identify this organism? (1 point)

b. Describe this organism's ecosystem importance? (2 points)

c. In what conditions do these organisms prefer to live in? (2 Points)

d. (T/F) The larvae are teleplanic (1 points)

4.

a. What is the common name of this organism? (1 point)

b. Describe the components of the skeleton of this organism. How does the fan-like shape of this organism aid feeding habits. (4 points)

c. Where do these organisms indicate unstable coral reefs? What type of conditions do these organisms prefer? (2 points)

5.

a. Identify the species (1 point)

b. **DIVISION C**: Name the life stage (1 point)

c. The channels leading through the body of the organism are called what? (2 points)

d. (T/F) These organisms are indicators of excessive nutrients

6.

a. Identify this organism (1 point)

b. When a fish of this species wants to become lighter, they will _____ their chromatophores. (2 points)

c. (T/F): Mature fish are apex predators

Water Monitoring and Analysis

1. What was your reading of salinity of the given sample in ppm? (**Tiebreaker**)

C. Tiebreaker

1. Red mangroves grow on the _____ side of a mangrove swamp and are salt _____. While white mangroves grown on the _____ side of a mangrove swamp and are salt _____.
    a. seaward, excluder, landward, extruder
    b. landward, excluder, seawardm extruder
    c. seaward, extruder, landward, excluder
    d. ladward, extruder, seaward, extruder
2. To keep the highest ratio of the most effective form of chloramine, aka _____, a pH of _____ should be kept.
    a. NHCl2, 6
    b. NH2Cl, 7
    c. NH2Cl, 6
    d. NHCl2, 7
3. Eutrophication of upstream rivers tends to decrease the availability of ____
    a. N
    b. O

c. S

d. Si

4. Under a pH of 6, _____ is inhibited

   a. Denitrification

   b. Ammonification

   c. Mineralization

   d. Nitrification

5. Most aquatic organisms are ____

   a. Eurytherms

   b. Stenotherms

   c. Homeotherm

   d. Poikilotherms